

Computer Vision Project — Final Report

Helmet Detection

YOLOv8n Object Detection on Traffic Images

Dataset: Traffic Laws Violations (Roboflow)

Item	Verified Project Detail
Model	YOLOv8n (nano)
Classes	Bike, Helmet, Passenger (3)
Total images	1,450
Train / Validation / Test	1,217 / 155 / 78
Image size	640 × 640 px
Training	50 epochs; early stopping configuration
Hardware	Tesla T4 GPU — Google Colab
Framework	Ultralytics YOLOv8

Report status: Corrected using the supplied final report and the supplied Colab notebook. Evaluation metrics are intentionally not fabricated because the notebook file contains no saved execution outputs for those values.

1. Project Overview

This project implements a real-time helmet detection system using the YOLOv8n object-detection architecture. The system identifies three classes — Bike, Helmet, and Passenger — in traffic surveillance images, with the objective of supporting detection of road-safety violations, particularly motorcycle riders who may not be wearing helmets.

The dataset is the Traffic Laws Violations dataset obtained from Roboflow. The project uses Google Colab and a Tesla T4 GPU for model training.

Objectives

- Train YOLOv8n to detect Bikes, Helmets, and Passengers.
- Evaluate the trained detector using standard object-detection metrics.
- Perform error analysis to identify model weaknesses.
- Document the complete workflow from data preparation through inference.

2. Dataset Description

The dataset contains 1,450 traffic images divided into training, validation, and test sets. The report records 1,217 training images, 155 validation images, and 78 test images.

Split	Images	Label files	Approx. share
Train	1,217	213	83.9%
Validation	155	155	10.7%
Test	78	78	5.4%
Total	1,450	—	100%

Important dataset limitation: only 213 of the 1,217 training images have label files according to the supplied report. Therefore, the effective annotated training set is substantially smaller than the total number of training images. This limitation should be disclosed during evaluation.

Classes: Bike (ID 0), Helmet (ID 1), Passenger (ID 2).

The existing dataset split is approximately 84% / 11% / 5%. This distribution is retained rather than altered in the corrected report.

3. Model Architecture

The project uses YOLOv8n, the nano variant of the Ultralytics YOLOv8 object-detection family. It is designed for relatively fast inference and resource-constrained environments.

Component	Project Detail
Model	YOLOv8n
Parameters	Approximately 3.0 million
GFLOPs	Approximately 8.2
Input	640 × 640 px
Pretraining	COCO
Fine-tuning	Helmet dataset; 3 classes

Transfer learning was used from the pretrained YOLOv8n model rather than training a detector from scratch. This is appropriate for a relatively small annotated dataset.

4. Training Configuration

Parameter	Value
Base model	yolov8n.pt
Epochs	50
Image size	640 px
Batch size	16
Optimizer	Auto / AdamW
Initial learning rate	0.01
Early stopping	Patience 15
AMP	True
Mosaic	1.0
Horizontal flip	0.5
HSV augmentation	h=0.015, s=0.7, v=0.4
Workers	2
Device	GPU:0 — Tesla T4

5. Model Evaluation

The project should report Precision, Recall, mAP@0.5, mAP@0.5:0.95, and per-class AP using the actual evaluation output generated by the Colab run.

Important: The supplied notebook does not contain saved execution output for these numerical metrics. Therefore, the corrected report leaves these fields as “Not available in supplied notebook” rather than inserting invented values.

Metric	Validation	Test
Precision	Not available in supplied notebook	Not available in supplied notebook
Recall	Not available in supplied notebook	Not available in supplied notebook
mAP@0.5	Not available in supplied notebook	Not available in supplied notebook
mAP@0.5:0.95	Not available in supplied notebook	Not available in supplied notebook

Per-class AP@0.5 values for Bike, Helmet, and Passenger are likewise not available in the supplied notebook outputs. These should be added after running the final validation/test evaluation.

6. Error Analysis

Expected error categories documented for this project include false positives, false negatives, wrong-class predictions, inaccurate bounding boxes, small-object failures, lighting problems, and annotation issues.

Error category	Typical project cause
False positive	A non-target object or shadow is classified as a target.
False negative	Small or occluded helmet/passenger/bike is missed.
Wrong class	An object is detected but assigned the wrong class.
Poor bounding box	Object is detected but localization is inaccurate.
Small-object failure	Distant helmets occupy very few pixels.
Lighting problem	Dark scenes or strong shadows reduce confidence.
Annotation issue	Incorrect or imprecise source labels can affect learning.

Note: The previous report contained example filenames such as image_01.jpg and placeholder predictions. They have been removed from this corrected version because the supplied notebook does not provide verifiable executed error-case outputs.

7. Limitations and Suggested Improvements

1. The main limitation is the small number of annotated training images: 213 label files are reported for 1,217 training images.
2. The dataset split is 84% / 11% / 5%, rather than an approximate 70% / 20% / 10% split.
3. YOLOv8n is optimized for speed but may have difficulty with very small helmets.
4. Low-light conditions and occlusion can reduce detection quality.
5. Quantitative video-inference performance was not evaluated in the supplied report.

Recommended improvements

- Annotate the remaining training images where appropriate.
- Experiment with YOLOv8s or YOLOv8m if computational resources permit.
- Add low-light and domain-specific augmentation.
- Increase the diversity and quality of training examples.
- Review class balance and annotation quality.
- Re-run evaluation after each major dataset/model change.

8. Project Structure and Submission

The final project structure documented in the supplied report is:

```
cv_project/  
■■■ dataset/  
■ ■■■ images/ (train, val, test)  
■ ■■■ labels/ (train, val, test)  
■ ■■■ data.yaml  
■■■ notebooks/helmet_detection_assignment.ipynb  
■■■ outputs/  
■■■ predictions/  
■■■ demo_results/  
■■■ models/best.pt  
■■■ report/final_report.pdf  
■■■ README.md  
■■■ requirements.txt
```

Requirements

ultralytics • torch • opencv-python • matplotlib • seaborn • pandas • pillow • pyyaml

9. Inference

The trained model can be loaded with Ultralytics YOLO and used for image or directory inference. The documented inference configuration uses a confidence threshold of 0.25 and an IoU threshold of 0.45.

```
from ultralytics import YOLO
model = YOLO('models/best.pt')

results = model.predict('your_image.jpg', conf=0.25, iou=0.45)
results[0].show()

results = model.predict('dataset/images/test/', conf=0.25)
```

10. Conclusion

This project implements an end-to-end computer-vision pipeline for detecting Bikes, Helmets, and Passengers in traffic images using YOLOv8n. The workflow covers dataset preparation, transfer learning, model training, evaluation planning, inference, and error analysis.

The most significant documented limitation is the availability of only 213 training label files for 1,217 training images. Improving annotation coverage is therefore a key next step. The project is suitable as a portfolio and academic computer-vision project, provided that the final numerical evaluation metrics are added from the actual executed Colab results before submission.

Final Submission Checklist

Requirement	Status
Total images ≥ 150 –300	Met — 1,450 images
Classes 2–5	Met — 3 classes
At least 100 annotated images	Met — 446 labeled files reported
YOLOv8n minimum	Met
Training epochs ≥ 30	Met — 50 epochs
Inference ≥ 15 test images	Reported as met — 20 images
Video inference	Optional / not quantitatively evaluated
Final report	Met
Actual evaluation metrics	Must be added from executed Colab output

Prepared from the supplied final report and helmet_detection_assignment.ipynb. September 11, 2026.